left side, the driver rotates the steering wheel according to the turning direction, at that time the load is acting on the axle shaft of the wheel which is going to turn. Hence the sun gear which is connected to it is also subjected to the load and reduces the rotation. At the same time the planet gears which are connected on the top and bottom of it rotate at high speed and rotate the sun gear which is located on other side. Hence the outside wheel rotates at higher speed than the inner wheel and turns the vehicle for certain distance easily. In this case the speed lost by the inner wheel is gained by the outer wheel and hence the vehicle turns easily on either left or right side.

4.10 Connecting methods of bevel gears

Based on the arrangement of the teeth in the pinion gear, the connecting methods with crown wheel are classified in to four different types.

1. Straight bevel gears
2. Spiral bevel gears
3. Hypoid gears
4. Worm and worm wheel

4.10.1 Straight bevel gears

In this type the gear teeth are cut in straight and connected at the center of the pinion. To operate this more amount of energy is required. In addition, when it is in operation, it produces more noise and it wears quickly. Figure 4.10.1 shows the view of straight bevel gear.



Figure 4.10.1

4.10.2 Spiral bevel gears

In this type of gears the teeth are cut in the inclined position. Hence the teeth are very well in mesh with the other teeth and the energy is transferred very softly and uniformly. In this type the axis of the crown wheel and the pinion are arranged in the same axis. This gear works without any noise and lasts for a long time. Figure 4.10.2 shows the view of spiral bevel gear arrangement.



Figure 4.10.2 Spiral bevel gear

Advantages

1. It has a strong arrangement.
2. It is more efficient.
3. Silent and smooth in operation.

BAE_ENG_Chapter_04.indd 62 12/10/2019 11:29:24 AM